

NurseMentor AI (Nightingale)

Problem Statement Name: NurseMentor AI (Nightingale)

Domain: Healthcare

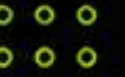
Team: TechCoder

R Naghanasri

**HACK
MATRIX**



The Problem



Competent,
Capable Nurses
with high
Problem-Solving

Recently
launched AI-tools
focus on content
for common
audience

AI tutor – Unable
to assess the
user's knowledge

Who are affected?

Nurses (Primary) -> Healthcare

Secondary -> Patient Health Impact

Problem Statement Name: NurseMentor AI (Nightingale)



Why this problem exists

Shortage of
Nursing
Instructors in
Rural Areas

Lack of
Clinical
Reasoning

Less or No
Personalized
Guidance

No Clinical real-life
Scenarios (Case
Studies)

– Nurse's Struggle in
Diagnostic confidence
and Clinical Reasoning

– Suboptimal Clinical
readiness & Increased
Clinical Risks

Current Approach:

Textbook Memorization → Lack of Feedback → Exam Anxiety →

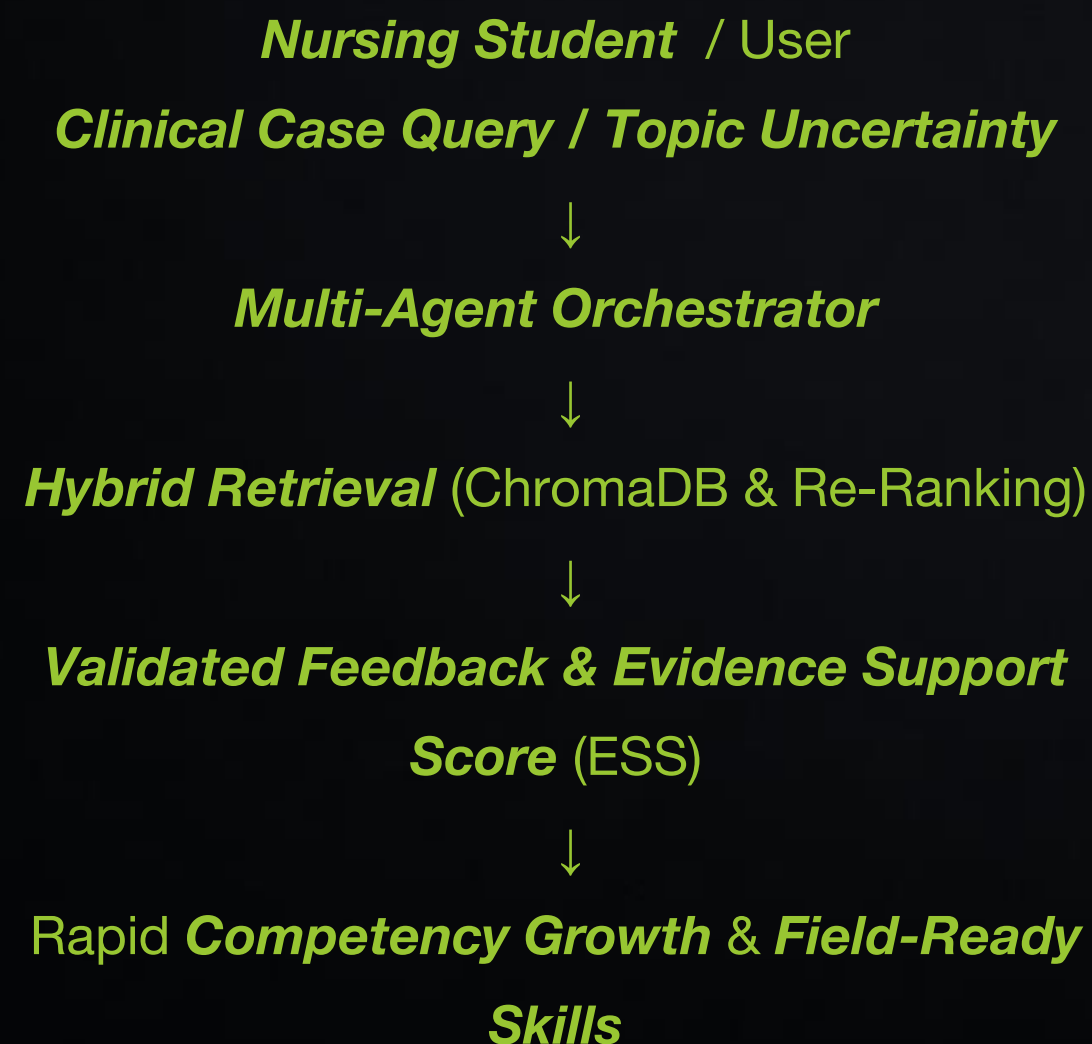
Underprepared Field Nurses



Our Solution

INTRODUCING *NurseMentor AI*.

NurseMentor AI enables Nurses to master Clinical Decision making by Multi-Agent Orchestration and evidence-based RAG.



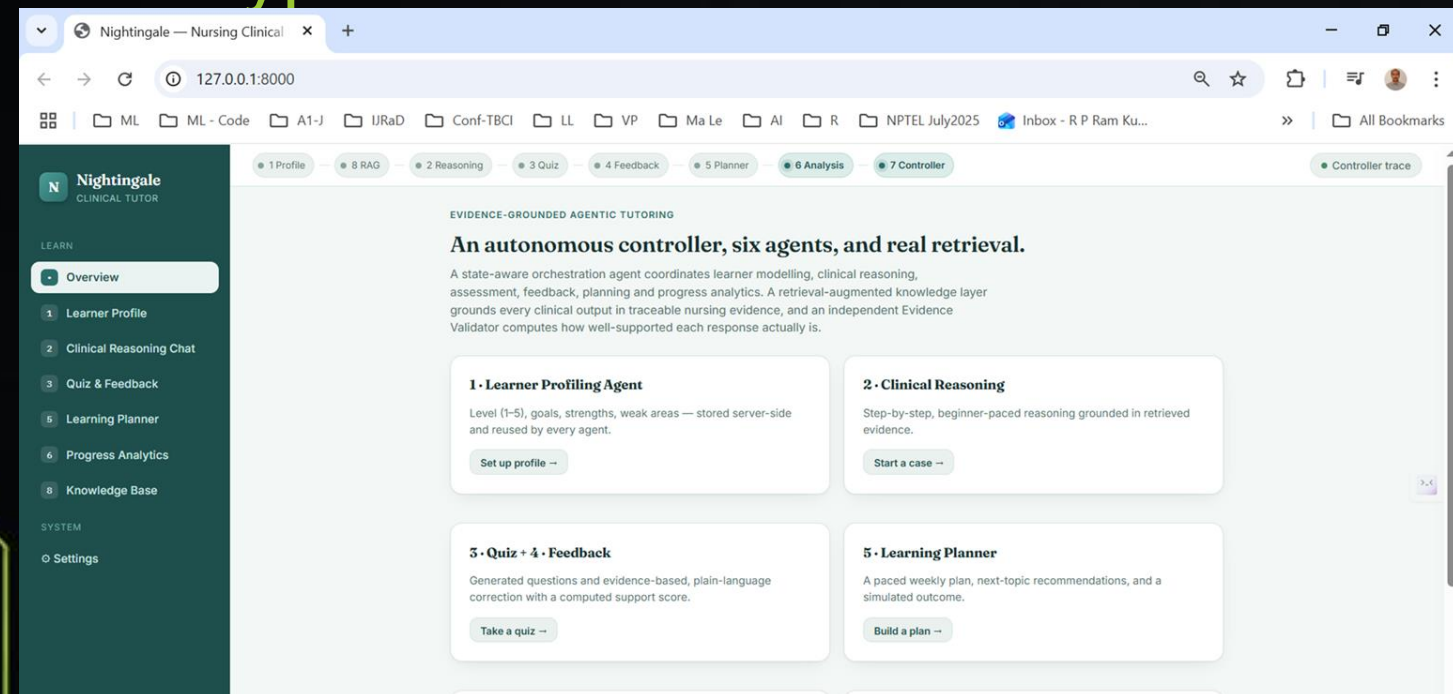
Core Capabilities:

- 01 **Orchestrator Agent** – Delivers Custom-made responses for every Query.
- 02 **Medical-Grade RAG** – Every answer provided will be strictly adhered to the *WHO Rulebook* & Hospital *Systematic Operating Procedures (SOP)*.
- 03 **Evidence Support Score (ESS)** – Verifies Factual Support, Detect contradictions & ensure Clinical Safety.

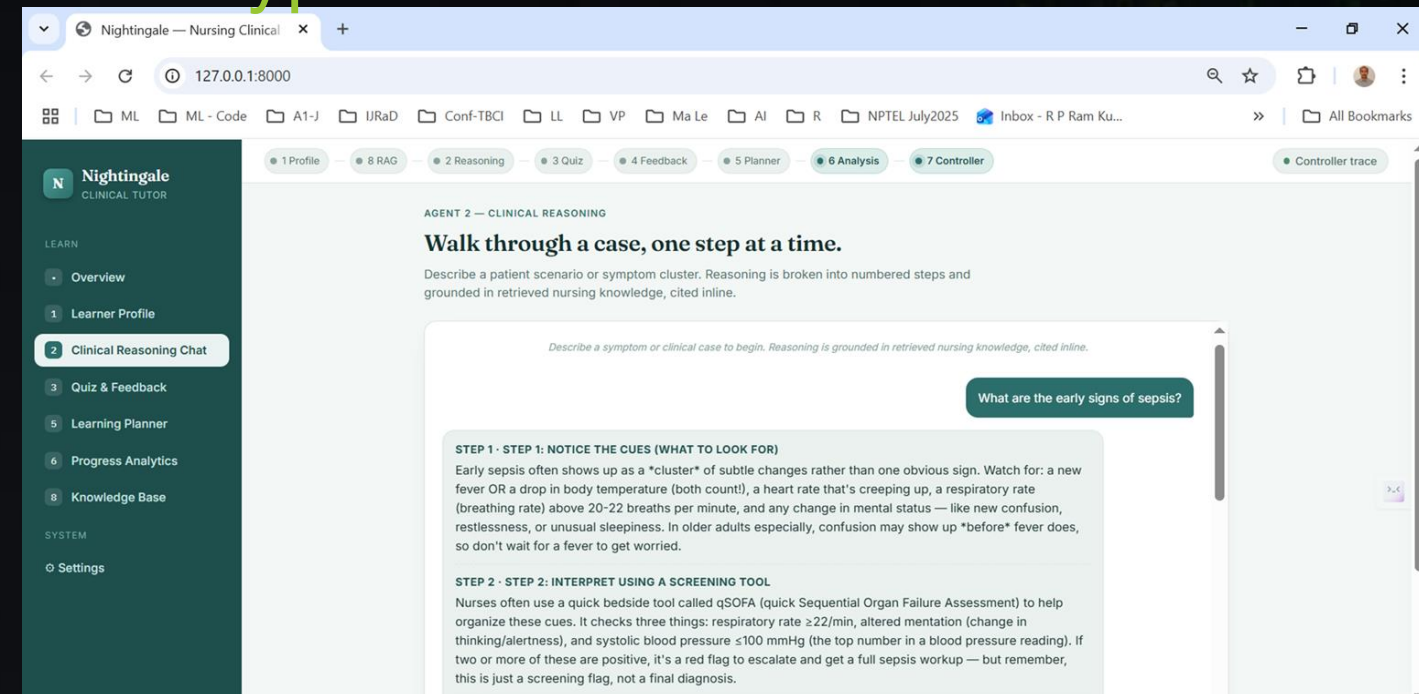


Prototype / Product Experience

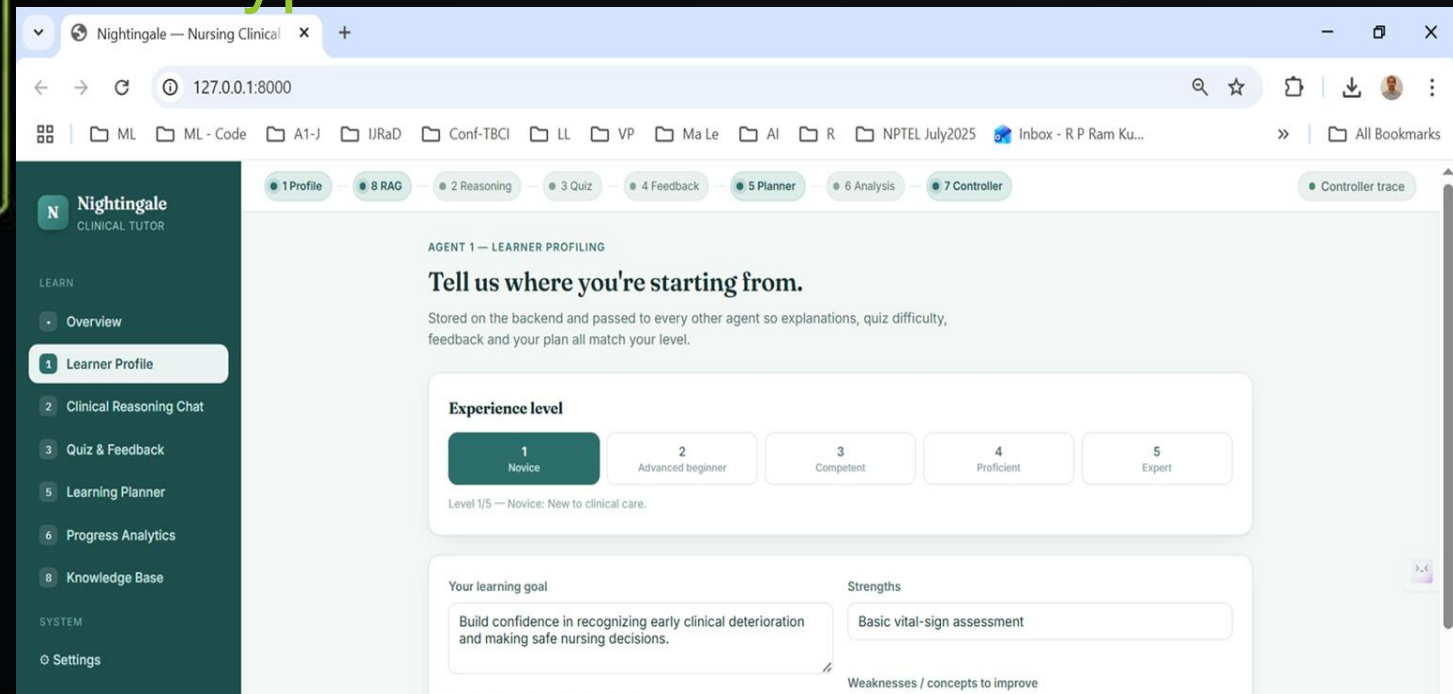
Prototype Screenshots 1



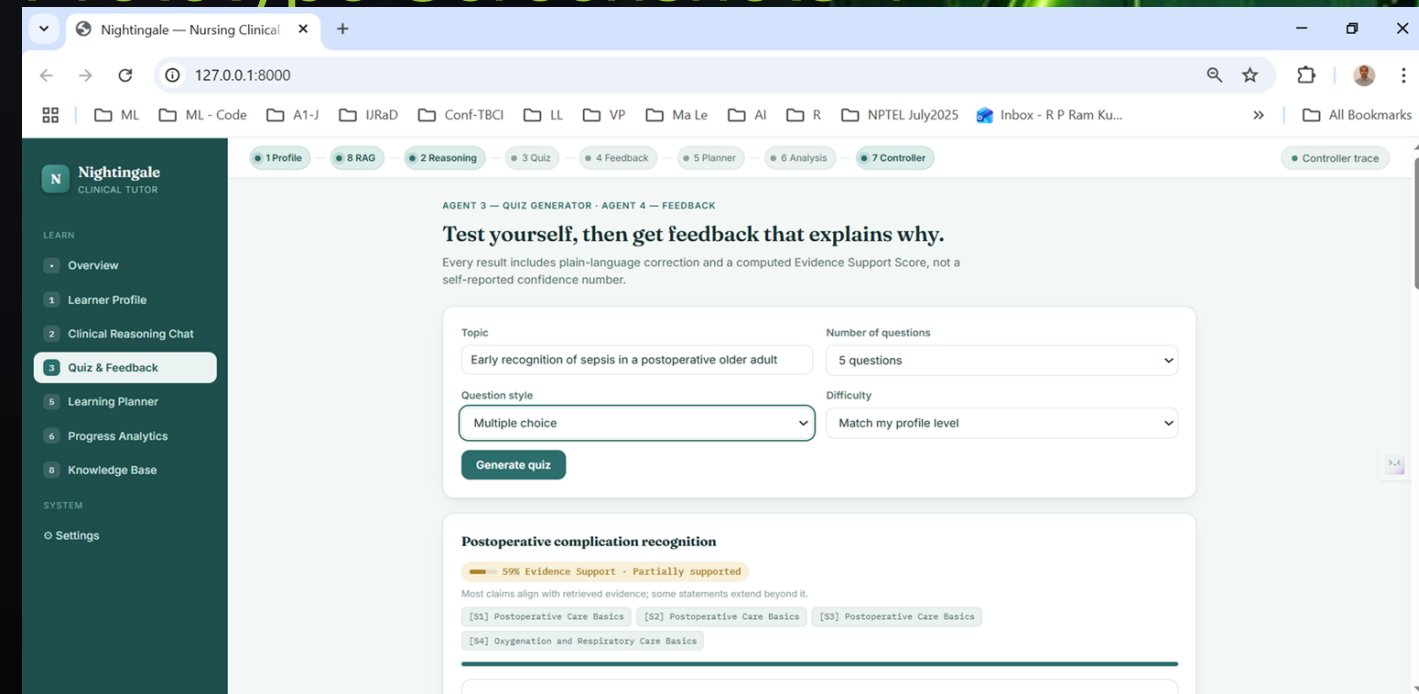
Prototype Screenshots 2



Prototype Screenshots 3



Prototype Screenshots 4



What makes us different ?

WHY THIS

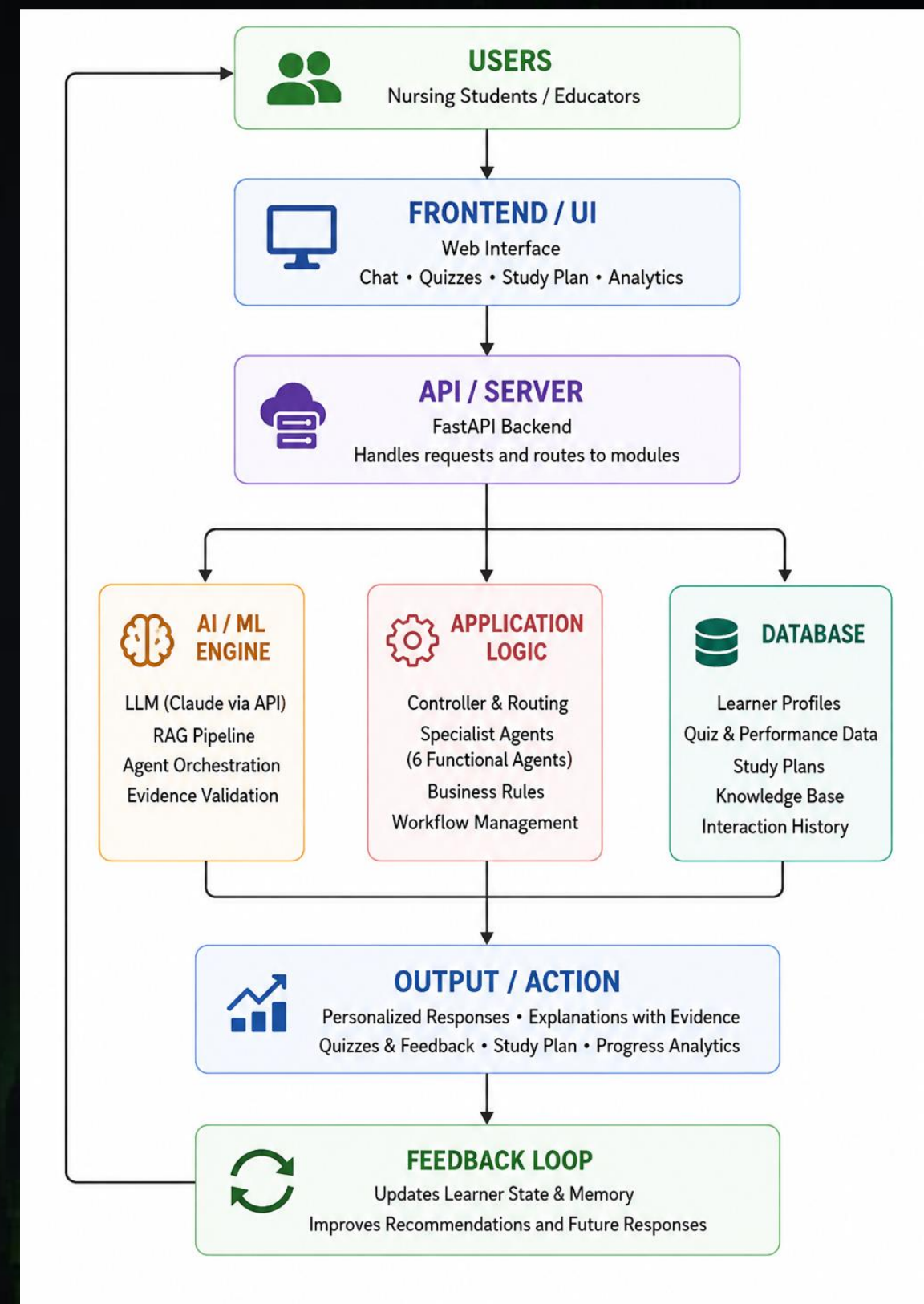
<i>Evaluation Approach</i> <i>Feature</i>	<i>Generic LLMs</i>	<i>NurseMentor AI</i>
<i>Response & Reliability</i>	High Risk of Hallucination (Unverified Answers)	Evidence-Grounded via RAG & Hybrid Search
<i>Safety & Validation</i>	None (Blind Text Generation)	Real-Time Safety Layer with Evidence Support Score (ESS)
<i>Architecture</i>	Single Monolithic Prompt / Basic Bot	Multi-Agent Orchestration (Specialized Duty Agents)
<i>Learning Workflow</i>	Passive Q&A or Static Textbooks	Adaptive Clinical Reasoning & Dynamic Scenario Generation
<i>Progress Tracking</i>	Static Quiz Scores	Real-Time Competency Growth & Automated Path Revision



Technical Architecture

Tech stack:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python, Fast API
- **AI Layer:** Large Language Model (LLM's) and Multi-Agent Architecture
- **Knowledge Layer:** Retrieval-Augmented Generation (RAG), Hybrid Retrieval and Re-ranking
- **Data Layer:** Learner profile, quiz history, progress and learning-state storage
- **Deployment:** Local web-based application / API-driven architecture



Feasibility + Impact

FEASIBILITY

- **Technical:** Claude LLM API, ChromaDB and RAG-based Retrieval.
- **Operational:** Standard Cloud Hosting, Open Source Medical Datasets & Standard API Endpoints.
- **Hackathon MVP:** Working on multi-agent workflow demonstrating a nursing student working through a clinical case, obtaining real-time RAG-grounded feedback, and evaluating the acquired knowledge through quiz.

IMPACT

Who benefits?

- Nurses (Novice to Expert Level)

What improves?

- Faster Clinical Decision Mastery when compared with Traditional Processing & Improvised Medical Accuracy Assurance

Why does it matter?

- Reduces Real-World Clinical Errors & trains Nurse for real-world Scenarios



Implementation Roadmap

WHAT HAPPENS NEXT?

NOW



NEXT



LATER



SCALE

- Core Agentic RAG Pipeline
- Grounding & Safety Agent
- Interactive Web-Interface

- Database Persistence
- Expansion of ChromaDB
- Re-Ranking Accuracy

- Upgrading to larger scale
- Educator Portals
- Multilingual Processing

- Expansion to healthcare centers
- Apt for remote support
- More specialized Cases



Team + Execution



Name: R Naghanasri

Role: Full-Stack AI Developer

Expertise: Python, RAG Pipelines,
Agentic Workflows and
Orchestration, Multi-Agent Systems

As an individual, integrated rapid prototyping expertise with orchestration, and deployment as a fully functional ***multi-agent clinical mentor***.

